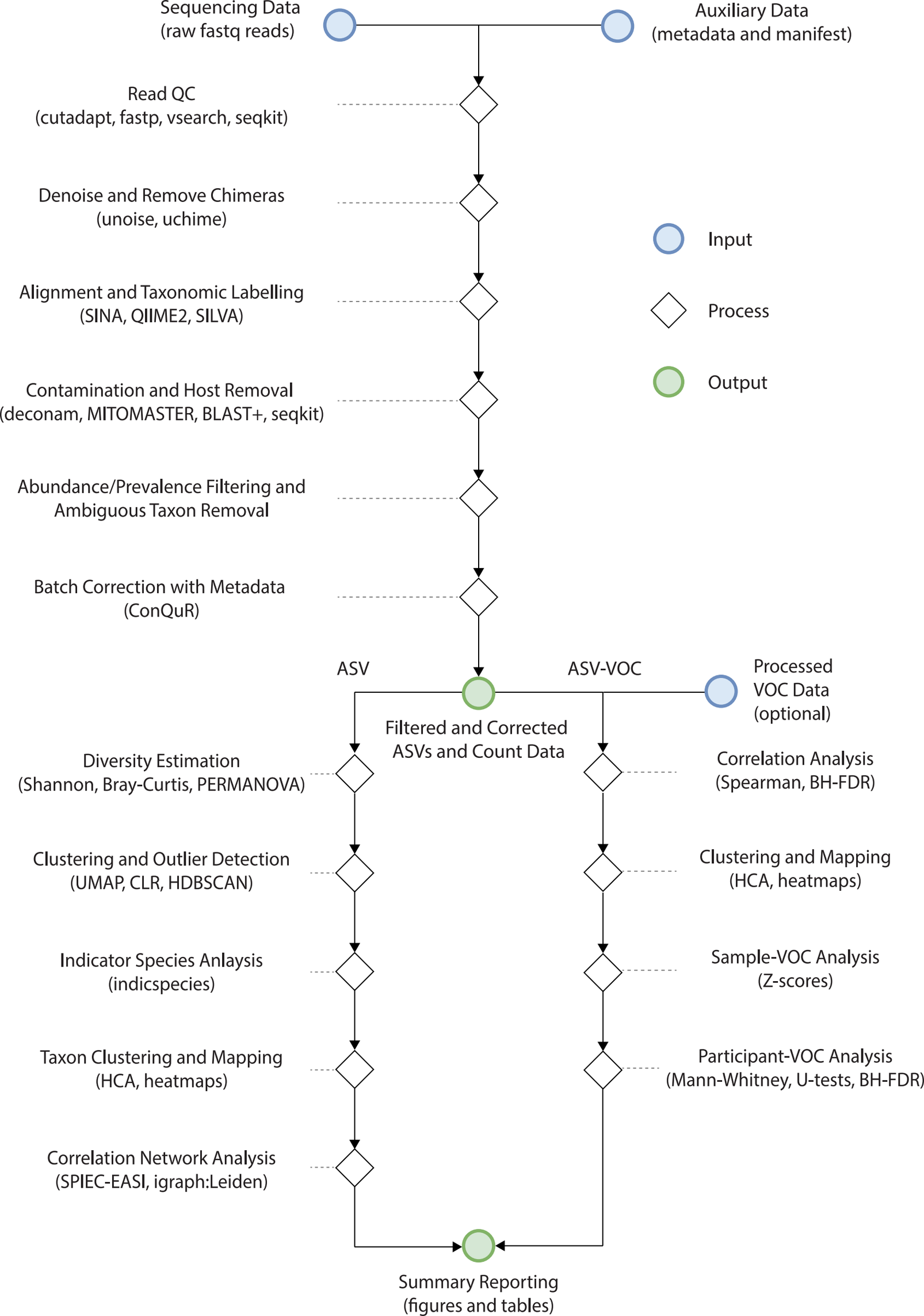

Sequencing Data
(raw fastq reads)
Auxiliary Data
(metadata and manifest)
Read QC
(cutadapt, fastp, vsearch, seqkit)
Denoise and Remove Chimeras
(unoise, uchime)
Alignment and Taxonomic Labelling
(SINA, QIIME2, SILVA)
Contamination and Host Removal
(deconam, MITOMASTER, BLAST+, seqkit)
Abundance/Prevalence Filtering and
Ambiguous Taxon Removal
Batch Correction with Metadata
(ConQuR)
Input
Process
Output
ASV
ASV-VOC
Processed
VOC Data
(optional)
Filtered and Corrected
ASVs and Count Data
Diversity Estimation
(Shannon, Bray-Curtis, PERMANOVA)
Correlation Analysis
(Spearman, BH-FDR)
Clustering and Outlier Detection
(UMAP, CLR, HDBSCAN)
Clustering and Mapping
(HCA, heatmaps)
Indicator Species Anlaysis
(indicspecies)
Sample-VOC Analysis
(Z-scores)
Taxon Clustering and Mapping
(HCA, heatmaps)
Participant-VOC Analysis
(Mann-Whitney, U-tests, BH-FDR)
Correlation Network Analysis
(SPIEC-EASI, igraph:Leiden)
Summary Reporting
(figures and tables)